

Smart Irrigation System using Soil Moisture Sensor in Tinkercad

Circuit Components:

1. Arduino UNO
2. Soil Moisture Sensor
3. DC Motor (Water Pump)
4. LED
5. Resistor (220 Ω)
6. Breadboard
7. 9V Battery
8. Jumper Wires

Arduino Code:

```
int sensorPin = A0;
int ledPin = 11;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int moisture = analogRead(sensorPin);

  Serial.println(moisture);

  if (moisture < 600) {
    digitalWrite(ledPin, HIGH);
  } else {
```

```
digitalWrite(ledPin, LOW);  
}  
  
delay(1000);  
}
```

```
1  int sensorPin = A0;  
2  int ledPin = 11;  
3  
4  void setup() {  
5      pinMode(ledPin, OUTPUT);  
6      Serial.begin(9600);  
7  }  
8  
9  void loop() {  
10     int moisture = analogRead(sensorPin);  
11  
12     Serial.println(moisture);  
13  
14     if (moisture < 600) {  
15         digitalWrite(ledPin, HIGH);  
16     } else {  
17         digitalWrite(ledPin, LOW);  
18     }  
19  
20     delay(1000);  
21 }
```

Circuit Connections:

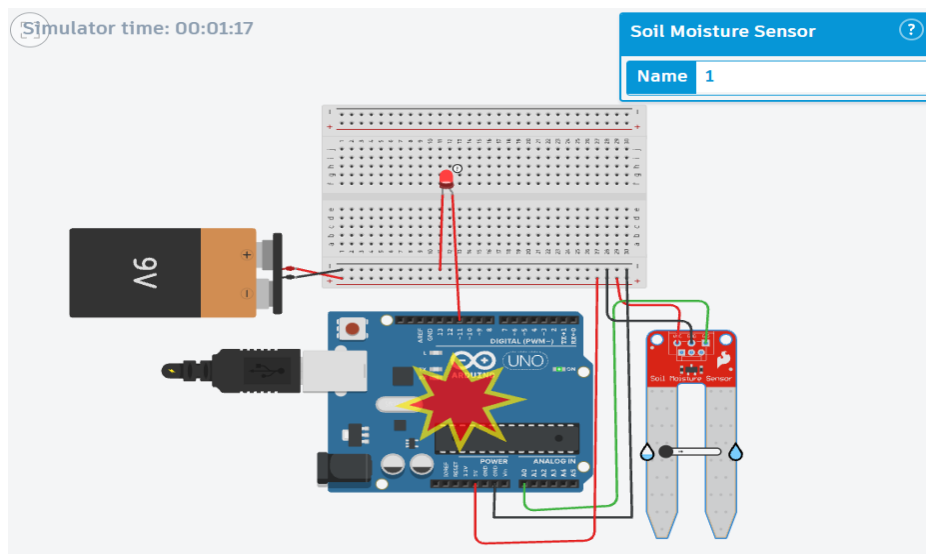
- Soil Moisture Sensor VCC → Arduino 5V
- Soil Moisture Sensor GND → Arduino GND
- Soil Moisture Sensor Signal → Arduino A0
- LED Anode (+) → Arduino Digital Pin
- LED Cathode (-) → GND through 220 Ω resistor
- DC Motor connected through Arduino output and external 9V supply
- Common GND connection maintained throughout the circuit

Working:

- The soil moisture sensor detects the moisture level in the soil.
- Arduino continuously reads the sensor value through pin A0.
- When the soil becomes dry, the sensor value falls below the threshold.
- Arduino turns ON the LED .
- When the soil becomes wet, Arduino turns OFF the LED.
- Thus, the system automatically irrigates the plant whenever required.

Output:

Dry Soil → LED ON.



Wet Soil → LED OFF.

Simulator time: 00:02:09

